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DISPLAY DEVICE, DISPLAY DEVICE CONNECTION SYSTEM, AND METHOD FOR DISPLAY DEVICE CONNECTION

Abstract

A display device, a display device connection system, and a method for display device connection are provided. The method includes the following steps. An audio signal including an identifier code is broadcasted by the display device. The audio signal is received by a terminal device to obtain the identifier code. A cloud server is accessed by the terminal device to obtain connection information corresponding to the identifier code. The terminal device is communicatively connected to the display device according to the connection information.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims the priority benefit of Taiwan application serial no. 113107659, filed on Mar. 4, 2024. The entirety of the above-mentioned patent application is hereby incorporated by reference herein and made a part of this specification.

BACKGROUND

Technical Field

[0002] The disclosure relates to a wireless communication technology, and in particular to a display device and a display device connection system that uses wireless communication technology as a method for display device connection.

Description of Related Art

[0003] Currently, there are many technologies for wireless communication products (e.g., wireless projection collaboration technology) that aim to enable display devices within the same space to establish communication connections with each other. Establishing the communication connection between display devices often relies on manual actions by the user. For example, to connect a terminal device held by the user with a display device in the space, the user needs to first activate the wireless local area network function of the terminal device and switch the terminal device to the same wireless local area network as the display device. Next, the terminal device needs to search for the display device within the wireless local area network and then establish a communication connection with the display device. During the process of establishing the communication connection, the user of the terminal device may also need to input authentication information. These steps are not only inconvenient but also pose a risk of exposing the authentication information.

[0004] The information disclosed in this Background section is only for enhancement of understanding of the background of the described technology and therefore it may contain information that does not form the prior art that is already known to a person of ordinary skill in the art. Further, the information disclosed in the Background section does not mean that one or more problems to be resolved by one or more embodiments of the disclosure was acknowledged by a person of ordinary skill in the art.

SUMMARY

[0005] A display device, a display device connection system, and a method for a display device connection are provided to establish a communication connection between a terminal device carried by a user and a display device set up in a space.

[0006] Other objectives and advantages of the disclosure may be further understood from the technical features disclosed in the disclosure.

[0007] To achieve one, a part, or all of the above objectives, or other objectives, an embodiment of a display device connection system in the disclosure includes a display device, a terminal device, and a cloud server. The cloud server is communicatively connected to the display device and the terminal device. The display device is used to broadcast an audio signal containing an identifier code. The terminal device is used to receive the audio signal to obtain the identifier code and to obtain connection information of the display device corresponding to the identifier code. The connection information is from the cloud server. The terminal device is used to communicatively connect to the display device according to the connection information.

[0008] In an embodiment of the disclosure, the display device transmits hardware feature information and the connection information to the cloud server. The cloud server is used to generate the identifier code according to the hardware feature information and to transmit the identifier code to the display device, wherein the identifier code corresponds to the connection

information.

[0009] In an embodiment of the disclosure, the hardware feature information includes at least one of a hardware serial number and a media access control address.

[0010] In an embodiment of the disclosure, the terminal device is communicatively connected to an electronic device. In response to the terminal device obtaining the identifier code of the display device, the terminal device is disconnected from the electronic device and is communicatively connected to the display device.

[0011] In an embodiment of the disclosure, the terminal device further includes a display screen, wherein the display screen is used to display a prompt screen to inquire about a user's selection.

[0012] In an embodiment of the disclosure, in response to the connection information being updated, the display device transmits the updated connection information to the cloud server.

[0013] In an embodiment of the disclosure, the audio signal includes an ultrasonic signal that is supported by sound transmission data technology, wherein the frequency of the ultrasonic signal is greater than 20,000 Hz.

[0014] In an embodiment of the disclosure, the connection information includes at least one of an Internet Protocol address and a service set identifier.

[0015] To achieve one, a part, or all of the above objectives, or other objectives, an embodiment of a method of a display device connection in the disclosure is applicable to a display device connection system including a display device, a terminal device, and a cloud server, wherein the cloud server is communicatively connected to the display device and the terminal device respectively. The method includes the following steps. The display device broadcasts an audio signal including an identifier code. The terminal device receives the audio signal to obtain the identifier code and accessing the cloud server to obtain connection information corresponding to the identifier code. The terminal device is communicatively connected to the display device according to the connection information.

[0016] In an embodiment of the disclosure, the display device transmits the hardware feature information and the connection information to the cloud server. The cloud server generates an identifier code according to the hardware feature information and transmits the identifier code to the display device, wherein the identifier code corresponds to the connection information.

[0017] In an embodiment of the disclosure, the hardware feature information includes at least one of a hardware serial number and a media access control address.

[0018] In an embodiment of the disclosure, the terminal device is communicatively connected to an electronic device. In response to the terminal device obtaining the identifier code of the display device, the terminal device is disconnected from the electronic device and is communicatively connected to the display device.

[0019] In an embodiment of the disclosure, the terminal device further includes a display screen, wherein the display screen displays a prompt screen to inquire about a user's selection.

[0020] In an embodiment of the disclosure, in response to the connection information being updated, the display device transmits the updated connection information to the cloud server.

[0021] In an embodiment of the disclosure, the audio signal includes an ultrasonic signal that is supported by sound transmission data technology, wherein the frequency of the ultrasonic signal is greater than 20,000 Hz.

[0022] In an embodiment of the disclosure, the connection information includes at least one of an Internet Protocol address and a service set identifier.

[0023] To achieve one, a part, or all of the above objectives, or other objectives, an embodiment of the display device in the disclosure includes a speaker, a communication device, and a processing device. The processing device is electrically connected to the speaker and the communication device. The processing device transmits the hardware feature information and the connection information via the communication device and is used to receive an identifier code corresponding to the hardware feature information. The processing device broadcasts an audio signal including the

identifier code via the speaker. The processing device is communicatively connected to the terminal device via the communication device according to the connection information.

[0024] Based on the above, the disclosure enables a terminal device carried by the user to be communicatively connected to a display device in a space, preventing the terminal device from connecting to an incorrect display device (e.g., a device is outside the space).

[0025] Other objectives, features and advantages of the present invention will be further understood from the further technological features disclosed by the embodiments of the present invention wherein there are shown and described preferred embodiments of this invention, simply by way of illustration of modes best suited to carry out the invention.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0026] FIG. 1 is a schematic diagram illustrating a display device connection system according to an embodiment of the disclosure.

[0027] FIG. 2 is a schematic diagram illustrating a method for display device connection according to an embodiment of the disclosure.

[0028] FIG. 3 is a schematic diagram illustrating a display device in spaces according to an embodiment of the disclosure.

[0029] FIG. 4 is a flowchart illustrating a method for display device connection according to an embodiment of the disclosure.

DESCRIPTION OF THE EMBODIMENTS

[0030] It is to be understood that other embodiment may be utilized and structural changes may be made without departing from the scope of the present invention. Also, it is to be understood that the phraseology and terminology used herein are for the purpose of description and should not be regarded as limiting. The use of “including,” “comprising,” or “having” and variations thereof herein is meant to encompass the items listed thereafter and equivalents thereof as well as additional items. Unless limited otherwise, the terms “connected,” “coupled,” and “mounted,” and variations thereof herein are used broadly and encompass direct and indirect connections, couplings, and mountings.

[0031] FIG. 1 is a schematic diagram illustrating a display device connection system **10** according to an embodiment of the disclosure, in which the display device connection system **10** may include a display device **100**, a terminal device **200**, and a cloud server **300**. The cloud server **300** may be communicatively connected to the display device **100**. The cloud server **300** may be communicatively connected to the terminal device **200**. The display device **100** is located in a space. For example, the display device **100** may be located in a conference room or classroom. The display device **100**, for example, is a projector or an interactive flat panel (IFP) display with touch functionality. The terminal device **200**, for example, is a mobile device carried by the user. The terminal device **200** may be a smartphone, tablet, or laptop computer.

[0032] The display device **100** may include a processing device **110**, a storage medium **120**, a communication device **130**, a speaker **140**, and a display screen **150**.

[0033] The processing device **110**, for example, includes at least one processor. The processor may be a central processing unit (CPU) or other programmable general-purpose or special-purpose microcontrol unit (MCU), microprocessor, digital signal processor (DSP), programmable controller, application-specific integrated circuit (ASIC), graphics processing unit (GPU), image signal processor (ISP), image processing unit (IPU), arithmetic logic unit (ALU), complex programmable logic device (CPLD), field-programmable gate array (FPGA), or other similar components, or a combination of the above components. The processing device **110** may be electrically connected to the storage medium **120**, the communication device **130**, the speaker **140**,

and the display screen **150**, respectively, and the processing device **110** may access and execute multiple applications stored in the storage medium **120**.

[0034] The storage medium **120**, for example, may be any type of fixed or removable random access memory (RAM), read-only memory (ROM), flash memory, hard disk drive (HDD), solid-state drive (SSD), or similar components, or a combination of the above components. The storage medium **120** is used to store multiple applications executable by the processing device **110**. In this embodiment, the storage medium **120** may store an application **21**. The processing device **110** may execute the application **21** to initiate various functions (e.g., connection function) of the display device **100**.

[0035] The communication device **130** transmits or receives signal wirelessly or via a wired connection. The communication device **130**, for example, may be a wireless network circuit or chip, a wired network circuit or chip, or a combination of these circuits or chips. In an embodiment, the communication device **130** may be a circuit or chip supporting Global System for Mobile Communication (GSM), Wireless Fidelity (WiFi), or Bluetooth communication technology, or a combination thereof, but is not limited thereto. The communication device **130** may also have specific circuits or chips for performing operations such as low-noise amplification, impedance matching, mixing, up or down frequency conversion, filtering, amplification, and similar functions. The processing device **110** of the display device **100** may be communicatively connected to the terminal device **200** or the cloud server **300** via the communication device **130**.

[0036] The terminal device **200** may include a processing device **210**, a storage medium **220**, a communication device **230**, an input interface **240**, a display screen **250**, and a sound receiving device **260**.

[0037] The processing device **210** of the terminal device **200** has similar hardware components to the processing device **110** of the display device **100** and will not be redundantly described here. The processing device **210** may be electrically connected to the storage medium **220**, the communication device **230**, the input interface **240**, the display screen **250**, and the sound receiving device **260**, respectively, and the processing device **210** may access and execute multiple applications stored in the storage medium **220**.

[0038] The storage medium **220** has similar hardware components to the storage medium **120** of the display device **100** and will not be redundantly described here. The storage medium **220** is used to store multiple applications executable by the processing device **210**. In this embodiment, the storage medium **220** may store an application **22**. The processing device **210** may execute the application **22** to initiate various functions (e.g., connection function) of the terminal device **200**.

[0039] The communication device **230** has similar hardware components to the communication device **130** of the display device **100** and will not be redundantly described here. The processing device **210** may be communicatively connected to the display device **100** or the cloud server **300** via the communication device **230**.

[0040] The processing device **210** may interact with the user via the display screen **250**. For example, the processing device **210** may display information (e.g., images) for the user to view on the display screen **250**. The input interface **240** may include a keyboard, buttons, or a touch device. The processing device **210** may receive commands from the user via the input interface **240**. The sound receiving device **260** may include audio receivers such as a dynamic microphone, condenser microphone, electret condenser microphone, micro-electrical mechanical system (MEMS) microphone, ribbon microphone, or carbon microphone. The sound receiving device **260** receives audio signal broadcasted by the speaker **140**, and the processing device **210** may receive the audio signal via the sound receiving device **260**.

[0041] Furthermore, the display screen **150** of the display device **100** and the display screen **250** of the terminal device **200** have similar hardware components. For example, the display screen **150** and the display screen **250** may be a liquid crystal display panel, LED display panel, micro LED display panel, or OLED display panel. The display screens **150** and **250** are used to display images.

[0042] The cloud server **300** may include a processing device **310**, a storage medium **320**, and a communication device **330**.

[0043] The processing device **310** has similar hardware components to the processing device **110** of the display device **100**, with differences in the quantity of hardware components and the speed and amount of data processing, which will not be redundantly described here. The processing device **310** may be electrically connected to the storage medium **320** and the communication device **330** and may access and execute multiple applications stored in the storage medium **320**.

[0044] The storage medium **320** has similar hardware components to the storage medium **120** of the display device **100**, with the difference being the amount of stored data. The storage medium **320** is used to store multiple applications executable by the processing device **310**.

[0045] The communication device **330** has similar hardware components to the communication device **130** of the display device **100** and will not be redundantly described here. The processing device **310** may be communicatively connected to the display device **100** or the terminal device **200** via the communication device **330**.

[0046] FIG. 2 is a schematic diagram illustrating a method for display device connection according to an embodiment of the disclosure, wherein the method may be implemented by the display device connection system **10** as shown in FIG. 1.

[0047] In step S201, the processing device **110** of the display device **100** executes the application **21** to allow the communication device **130** to transmit or upload information of at least one hardware feature and connection information to the cloud server **300** via the Internet. The hardware feature information is hardware information associated with the display device **100**; for example, the hardware feature information may at least include a hardware serial number or a media access control (MAC) address of the display device **100**. The connection information may include information necessary for connecting to the display device **100**, such as an Internet Protocol (IP) address, service set identifier (SSID), or password. For instance, if the display device **100** is connected to other terminal devices via a wireless network provided by a wireless router set up in the space, then the connection information may include the SSID and/or connection password of that wireless router. In another example, if the display device **100** itself has the functionality of a wireless router, then the connection information may include the SSID and/or connection password of the display device **100**. The above-mentioned hardware feature information and connection information may be stored in the storage medium **120** of the display device **100**.

[0048] In an embodiment, when the connection information of the display device **100** is updated, the application **21** executed by the processing device **101** of the display device **100** may transmit the updated connection information to the cloud server **300** in real time or periodically. For example, if the space where the display device **100** is located includes a first wireless router and a second wireless router, and the connection of the display device **100** switches from the first wireless router to the second wireless router, then the application **21** executed by the processing device **101** may update the connection information corresponding to the first wireless router to the connection information corresponding to the second wireless router. The application **21** executed by the processing device **101** may further transmit the updated connection information to the cloud server **300**.

[0049] After the cloud server **300** receives the hardware feature information and the connection information from the display device **100**, in step S202, the processing device **310** of the cloud server **300** may generate an identifier code based on the hardware feature information of the display device **100** and transmit the identifier code to the display device **100** via the communication device **330**. The identifier code corresponds to the connection information of the display device **100**, meaning that the connection information of the display device **100** may be obtained based on the identifier code. Through the communication device **130**, the display device **100** obtains the identifier code corresponding to the hardware feature information of the display device **100**. The identifier code is stored in the storage medium **320** of the display device **100**. The cloud server **300**

may store the identifier code and the connection information of the display device **100** in the storage medium **320**; additionally, the hardware feature information of the display device **100** may be also stored in the storage medium **320**.

[0050] In step **S203**, in the space, with the display device **100** in an active state, the processing device **110** of the display device **100** controls the speaker **140**, and the speaker **140** continuously or periodically broadcasts an audio signal containing the identifier code. At this time, the terminal device **200** operated by the user receives the audio signal containing the identifier code via the sound receiving device **260**, and the application **22** executed by the processing device **210** decodes the audio signal to obtain the identifier code. The audio signal may be an ultrasound signal supporting data over sound technology. For example, the processing device **110** of the display device **100** executes the application **21** to encode the identifier code as an audio barcode, the sound receiving device **260** receives the audio signal containing the identifier code (audio barcode), and the application **22** executed by the processing device **210** decodes the audio signal to obtain the identifier code. The frequency of the ultrasound signal may be greater than 20,000 Hz. Ultrasound signal is generally inaudible to humans. Ultrasound signal has low penetration characteristics, meaning they do not easily pass through physical objects, such as walls or partitions. Only terminal devices **200** within the same space as the display device **100** may receive the ultrasound signal and obtain the identifier code corresponding to the display device **100** from the ultrasound signal.

[0051] In step **S204**, the processing device **210** of the terminal device **200** executes the application **22** to establish a communication connection to the cloud server **300** via the Internet or a mobile network and uploads the identifier code to the cloud server **300**. The cloud server **300** is used to determine whether there is connection information for the display device **100** corresponding to the identifier code. Specifically, if none exists, the name of the display device **100** will not be displayed on the display screen **250**. If it exists, the name of the display device **100** is displayed on the display screen **250**. In step **S205**, if the cloud server **300** confirms the existence of connection information for the display device **100** corresponding to the identifier code, the cloud server **300** transmits the connection information of the display device **100** to the terminal device **200** via the Internet or a mobile network, and the application **22** executed by the processing device **210** may receive the connection information of the display device **100** from the cloud server **300**. In step **S206**, the application **22** executed by the processing device **210** may establish a communication connection to the display device **100** based on the connection information of the display device **100**. Specifically, the application **22** executed by the processing device **210** of the terminal device **200** may establish a communication connection with the communication device **130** of the display device **100** via the communication device **230**. In one example, both the terminal device **200** and the display device **100** are within the same wireless network domain, allowing the terminal device **200** to control the display device **100** so that the screen displayed on the display screen **250** of the terminal device **200** is projected onto the display screen **150** of the display device **100**, making the screen on display screen **250** identical to the screen on display screen **150**.

[0052] Further explanation: In step **S201**, the processing device **110** of the display device **100** executes the application **21**, enabling the communication device **130** to establish a communication connection with the cloud server **300** via the Internet. The location of the cloud server **300**, for example, may be in a nearby space or country relative to the display device **100**. For instance, if the location of the display device **100** is in Japan, when the display device **100** is connected to the cloud server **300** via the Internet, it will connect to the cloud server **300** in Taiwan (the cloud server of the specific region) rather than to a cloud server in Europe or the America. The hardware feature information and connection information of the display device **100** are stored in the cloud server **300** in Taiwan. It is worth mentioning that the cloud servers in each region run an information synchronization program at regular intervals to synchronize the data stored on the cloud server in one region with other cloud servers. Therefore, when the user executes the application **22** on the terminal device **200**, after logging in with the user's account and password, the terminal device **200**

connects to the cloud server **300** via the Internet or a mobile network and uploads the identifier code to the cloud server **300**. The cloud server **300** may then identify and provide the connection information of the display device **100** corresponding to the identifier code.

[0053] When the user initiates the execution of the application **22** on the terminal device **200** by the processing device **210**, after step **S205** is completed, a prompt screen is displayed on the display screen **250** of the terminal device **200**. For example, a prompt screen stating “Display device **100** detected and available for communication connection. Would you like to connect to display device **100**?” appears. The display screen **250** shows the names of at least one display device available for communication connection, allowing the user to select one of the available display devices. When the user selects the display device **100**, the terminal device **200** may establish a communication connection with the display device **100**.

[0054] In an embodiment, after the terminal device **200** completes a communication connection with the display device **100**, the application **22** executed by the processing device **210** of the terminal device **200** may disconnect the communication connection between the terminal device **200** and other electronic devices. Taking FIG. **3** as an example, suppose the user is carrying the terminal device **200** within a space **32**, and the terminal device **200** is communicatively connected to an electronic device **500**. The electronic device **500** in the space **32**, for instance, may have the same equipment and functionality as the display device **100**. When the user carries the terminal device **200** out of the space **32** and into another space **31**, the terminal device **200** receives the audio signal from the display device **100** and obtains the identifier code of the display device **100** from the audio signal. After obtaining the identifier code of the display device **100**, the terminal device **200** may disconnect from the electronic device **500** and establish a communication connection with the display device **100** based on the identifier code. Further explanation: when the application **22** executed by the processing device **210** of the terminal device **200** is running, it continuously or periodically receives the audio signal. The application **22** executed by the processing device **210** is used to confirm whether the sound receiving device **260** has received the latest audio signal.

[0055] In an embodiment, upon receiving the latest audio signal, the display screen **250** of the terminal device **200** displays the prompt screen as described above. After the user selects the new display device, the terminal device **200** may establish a communication connection with the new display device.

[0056] FIG. **4** is a flowchart illustrating a method for display device connection according to an embodiment of the disclosure, wherein the method may be implemented by the display device connection system **10** as shown in FIG. **1**. In step **S401**, a display device broadcasts an audio signal including an identifier code. In step **S402**, a terminal device receives the audio signal to obtain the identifier code, and accesses a cloud server to obtain connection information corresponding to the identifier code. The connection information is the connection information of the display device. In step **S403**, the terminal device is communicatively connected to the display device according to the connection information.

[0057] In summary, the display device, display device connection system, and display device connection method of embodiments in the disclosure have at least one of the following advantages: a display device set up in a space may broadcast an audio signal containing an identifier code, where the audio signal, for example, includes an ultrasound signal that supports data-over-sound technology and has a low penetration characteristic. Only terminal devices within the same space may receive the identifier code carried by the audio signal. The terminal device may use the identifier code to connect to the cloud server, thereby obtaining the connection information of the display device, which may contain a larger amount of information, through a mobile network or a wireless network. The terminal device may establish a communication connection with the display device based on the connection information. The terminal device carried by the user will only connect to a display device located in the same space. Accordingly, the disclosure improves the

communication connection functionality of the terminal device, preventing it from connecting to an incorrect target (e.g., a device locates outside the space) when performing the communication connection function. Additionally, the terminal device may quickly obtain the connection information from the cloud server through a mobile network or wireless network.

[0058] The foregoing description of the preferred embodiments of the invention has been presented for purposes of illustration and description. It is not intended to be exhaustive or to limit the invention to the precise form or to exemplary embodiments disclosed. Accordingly, the foregoing description should be regarded as illustrative rather than restrictive. Obviously, many modifications and variations will be apparent to practitioners skilled in this art. The embodiments are chosen and described in order to best explain the principles of the invention and its best mode practical application, thereby to enable persons skilled in the art to understand the invention for various embodiments and with various modifications as are suited to the particular use or implementation contemplated. It is intended that the scope of the invention be defined by the claims appended hereto and their equivalents in which all terms are meant in their broadest reasonable sense unless otherwise indicated. Therefore, the term “the invention”, “the present invention” or the like does not necessarily limit the claim scope to a specific embodiment, and the reference to particularly preferred exemplary embodiments of the invention does not imply a limitation on the invention, and no such limitation is to be inferred. The invention is limited only by the spirit and scope of the appended claims. Moreover, these claims may refer to use “first”, “second”, etc. following with noun or element. Such terms should be understood as a nomenclature and should not be construed as giving the limitation on the number of the elements modified by such nomenclature unless specific number has been given. The abstract of this disclosure is provided to comply with the rules requiring an abstract, which will allow a searcher to quickly ascertain the subject matter of the technical disclosure of any patent issued from the disclosure. It is submitted with the understanding that it will not be used to interpret or limit the scope or meaning of the claims. Any advantages and benefits described may not apply to all embodiments of the invention. It should be appreciated that variations may be made in the **10** embodiments described by persons skilled in the art without departing from the scope of the present invention as defined by the following claims. Moreover, no element and component in the present disclosure is intended to be dedicated to the public regardless of whether the element or component is explicitly recited in the following claims.

Claims

1. A display device connection system, comprising: a display device; a terminal device; and a cloud server, communicatively connected to the display device and the terminal device, wherein the display device is configured to broadcast an audio signal comprising an identifier code; the terminal device is configured to receive the audio signal to obtain the identifier code, and to obtain connection information of the display device corresponding to the identifier code configured, wherein the connection information is from the cloud server, and the terminal device is communicatively connected to the display device according to the connection information.
2. The display device connection system according to claim 1, wherein the display device is configured to transmit a hardware feature information and the connection information to the cloud server; and the cloud server is configured to generate the identifier code according to the hardware feature information and transmit the identifier code to the display device, wherein the identifier code corresponds to the connection information.
3. The display device connection system according to claim 2, wherein the hardware feature information comprises at least one of a hardware serial number and a media access control address.
4. The display device connection system according to claim 1, wherein the terminal device is communicatively connected to an electronic device; and in response to the terminal device obtaining the identifier code of the display device, the terminal device is communicatively

disconnected from the electronic device, and the terminal device is communicatively connected to the display device.

5. The display device connection system according to claim 1, wherein the terminal device further comprises a display screen, and the display screen is configured to display a prompt screen to inquire a selection of a user.

6. The display device connection system according to claim 1, wherein in response to the connection information being updated, the display device is configured to transmit the updated connection information to the cloud server.

7. The display device connection system according to claim 1, wherein the audio signal comprises an ultrasonic signal, wherein a frequency of the ultrasonic signal is greater than 20,000 Hz.

8. The display device connection system according to claim 1, wherein the connection information comprises at least one of an Internet Protocol address and a service set identifier.

9. A method of a display device connection, applicable to a display device connection system comprising a display device, a terminal device, and a cloud server, wherein the cloud server is communicatively connected to the display device and the terminal device respectively, the method comprising: broadcasting an audio signal comprising an identifier code by the display device; receiving the audio signal to obtain the identifier code, and accessing the cloud server to obtain connection information corresponding to the identifier code by the terminal device; and communicatively connecting the terminal device and the display device according to the connection information.

10. The method according to claim 9, further comprising: transmitting a hardware feature information and the connection information to the cloud server by the display device; and generating the identifier code according to the hardware feature information by the cloud server, and transmitting the identifier code to the display device, wherein the identifier code corresponds to the connection information.

11. The method according to claim 10, wherein the hardware feature information comprises at least one of a hardware serial number and a media access control address.

12. The method according to claim 9, further comprising: communicatively connecting the terminal device and an electronic device; and in response to the terminal device obtaining the identifier code of the display device, disconnecting the terminal device and the electronic device, and communicatively connecting the terminal device and the display device.

13. The method according to claim 9, wherein the terminal device further comprises a display screen, wherein the display screen is configured to display a prompt screen to inquire a selection of a user.

14. The method according to claim 9, further comprising: in response to the connection information being updated, transmitting the updated connection information to the cloud server by the display device.

15. The method according to claim 9, wherein the audio signal comprises an ultrasonic signal, wherein a frequency of the ultrasonic signal is greater than 20,000 Hz.

16. The method according to claim 9, wherein the connection information comprises at least one of an Internet Protocol address and a service set identifier.

17. A display device, comprising: a speaker; a communication device; and a processing device, electrically connected to the speaker and the communication device, wherein the processing device is configured to transmit hardware feature information and connection information through the communication device, and to receive an identifier code corresponding to the hardware feature information; the processing device is configured to broadcast an audio signal comprising the identifier code through the speaker; and the processing device is communicatively connected to a terminal device through the communication device according to the connection information.
